Incident Investigation Report

Document No: BPL-IIR-2024-0023

Date of Incident	2024-08-14 14:30	Location	Heat Exchanger E-103, CDU
Incident Type	Hydrocarbon Leak	Severity	Medium

Description of Event

During routine operator rounds, a minor naphtha leak (~2 liters/min) was observed from the tube-side flange of heat exchanger E-103. Area was immediately barricaded, and the bypass valve was opened to isolate E-103. Unit throughput was reduced by 10% during the isolation process.

Root Cause Analysis

Investigation revealed that the spiral wound gasket (SS316/Graphite) had degraded. The root cause was determined to be improper torqueing during the last maintenance turnaround, leading to uneven compression and eventual failure due to thermal cycling.

Corrective Actions

1. Replace gasket and torque to specification using a calibrated hydraulic torque wrench (Completed).
2. Revise maintenance procedure to mandate QA/QC sign-off on torque values for all Class 300 flanges and above.